

9 Magnesium

Magnesium sits at the center of chlorophyll and supports many enzymes. It is held on soil exchange sites, moves within the plant, and competes with other positively charged nutrients.

Why it matters

- **Photosynthesis.** Magnesium is the central atom in chlorophyll and is essential for capturing light energy.
- **Enzymes and transport.** It activates enzymes and supports energy transfer and movement of sugars.

Tentative leaf guide

Measure	Interpretation note
Tentative avocado range	UC guidance lists less than 0.15 percent as deficient, 0.25 to 0.80 percent as adequate, and more than 1.0 percent as excess.
Plant pattern	Because Mg is mobile in the plant, deficiency usually appears first on older leaves as interveinal yellowing.

Main soil and environmental controls

Factor	What it can mean in the root zone
pH and leaching	Acid sandy soils can lose Mg by leaching. Liming choices should be based on pH and whether Ca, Mg, or both are needed.
CEC and organic matter	Clay and organic matter hold Mg on exchange sites. Low-CEC soil has less reserve and may show larger swings.
Irrigation water	Water can be a meaningful Mg source. Include it in the annual budget and in salinity and hardness interpretation.
Cation competition	High K, Ca, ammonium, or Na can reduce Mg uptake in some settings. Sufficiency and response matter more than a fixed universal ratio.
Roots and moisture	Drought, salinity, waterlogging, compaction, and root disease reduce uptake even when exchangeable Mg is present.

Field clues and checks

- **Interpret the cation set.** Review Mg with K, Ca, Na, CEC, pH, soil texture, water chemistry, and fertilizer history.
- **Choose source by purpose.** Dolomitic lime also raises pH; magnesium sulfate also adds sulfate and EC. Neither is interchangeable with a root-zone correction.
- **Check old leaves and roots.** Separate Mg-like chlorosis from root stress, salinity, leaf age, and other nutrient patterns.

DST interpretation Phase 1 found a continuous nutrient-state pattern in which Mg moved with Ca, Mn, and Fe and opposite K, P, Zn, and B. This is useful context, not a fixed grove class or target ratio.